

|                                                    |                                                       |
|----------------------------------------------------|-------------------------------------------------------|
| Program : <b>Diploma in Information Technology</b> |                                                       |
| Course Code : <b>5267</b>                          | Course Title: <b>Network Routing Technologies Lab</b> |
| Semester : <b>5</b>                                | Credits: <b>1.5</b>                                   |
| Course Category: <b>Program Core</b>               |                                                       |
| Periods per week: <b>3 (L:0 T:0 P:3)</b>           | Periods per semester: <b>45</b>                       |

### Course Objectives:

- To provide practical exposure to Network Routing Technologies from a Network Administrator's point-of-view.
- To provide the basics of Network Router and its Administration.

### Course Prerequisites:

| Topic                                | Course Code | Course Title                           | Semester |
|--------------------------------------|-------------|----------------------------------------|----------|
| Subnetting, WAN, IP Address, Routing |             | Fundamentals of Network Administration | 3        |
| IOS basics, VLAN                     |             | Network Switching Technologies         | 4        |

### Course Outcomes

On completion of the course, students will be able to:

| COn | Description                                                                    | Duration (Hours) | Cognitive Level |
|-----|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1 | Experiment with Static Routing and RIP                                         | 10               | Applying        |
| CO2 | Experiment with VLAN and do inter-VLAN routing by using a router and L3 Switch | 10               | Applying        |
| CO3 | Experiment with EIGRP, OSPF and eBGP for IPv4 Network                          | 10               | Applying        |
| CO4 | Experiment with Static Routing, EIGRP and OSPF for IPv6 Network                | 12               | Applying        |
|     | Lab Exam                                                                       | 3                |                 |

## CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 3   |     |     | 3   |     |     |     |
| CO2             | 3   |     |     | 3   |     |     |     |
| CO3             | 3   |     |     | 3   |     |     |     |
| CO4             | 3   |     |     | 3   |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline

| Module Outcomes | Description                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Experiment with Static Routing and RIP                                         |                  |                 |
| M1.01           | Experiment with Router Basic Commands                                          | 4                | Applying        |
| M1.02           | Configure Static Routing on Routers                                            | 3                | Applying        |
| M1.03           | Configure RIP Routing on Routers                                               | 3                | Applying        |
| CO2             | Experiment with VLAN and do inter-VLAN routing by using a router and L3 Switch |                  |                 |
| M2.01           | Configure a VLAN                                                               | 3                | Applying        |
| M2.02           | Configure Inter-VLAN Routing using Router                                      | 3                | Applying        |
| M2.03           | Configure Inter-VLAN Routing using Layer 3 Switch                              | 4                | Applying        |
|                 | Lab Exam – I                                                                   | 1.5              |                 |
| CO3             | Experiment with EIGRP, OSPF and eBGP for IPv4 Network                          |                  |                 |
| M3.01           | Configure EIGRP                                                                | 3                | Applying        |
| M3.02           | Configure OSPF                                                                 | 3                | Applying        |
| M3.03           | Configure eBGP                                                                 | 4                | Applying        |
| CO4             | Experiment with Static Routing, EIGRP and OSPF for IPv6 Network                |                  |                 |
| M4.01           | Configure Static Routing for IPv6                                              | 2                | Applying        |
| M4.02           | Configure EIGRP for IPv6                                                       | 2                | Applying        |

|       |                           |     |          |
|-------|---------------------------|-----|----------|
| M4.03 | Configure OSPF for IPv6   | 2   | Applying |
| M4.04 | Open-ended experiment. ** | 6   | Applying |
|       | Lab Exam – II             | 1.5 |          |

### **\*\* Suggested Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Create a network model to interconnect all Polytechnic Colleges in Kerala using Packet Tracer and apply each routing protocols.

### **Text / Reference:**

| <b>T/R</b> | <b>Book Title / Author</b>                                                                                                                                        |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T1         | Todd Lammle, <i>CCNA Routing and Switching Complete Study Guide</i> , 2 <sup>nd</sup> Edition, Sybex (Wiley) Publication, 2016                                    |
| R1         | Narasimha Karumanchi, Dr. A. Damodaram and Dr. M. Sreenivasa Rao. <i>Elements of Computer Networking: An Integrated Approach</i> . CareerMonk Publications, 2014. |

### **Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/">https://www.freeccnastudyguide.com/study-guides/icnd2-200-101/</a>                                       |
| 2            | <a href="https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/">https://www.thebryantadvantage.com/videos-and-tutorials/ccna-and-ccent-tutorials/</a> |
| 3            | <a href="https://www.guru99.com/introduction-ccna.html">https://www.guru99.com/introduction-ccna.html</a>                                                                         |
| 4            | <a href="https://study-ccna.com/">https://study-ccna.com/</a>                                                                                                                     |
| 5            | <a href="http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php">http://www.omnisecu.com/cisco-certified-network-associate-ccna/index.php</a>                   |